Auditory superstitious-perception task: a focused report on Full-vs-Imagined block, site, and block-order effects

Undergraduate-thesis companion analysis

Abstract

This report distils the auditory superstitious-perception dataset ($N = 26$ subjects: 16 Oberlin, 10 UChicago, two 150-trial blocks each) to the three questions the thesis is built on: (i) does behaviour differ between the `full_sentence` and `imagined_sentence` blocks; (ii) do the two recruitment sites differ; and (iii) does block-order matter. Every analysis is repeated three times (UChicago alone, Oberlin alone, and unified pool) so that any interpretation can be cross-checked at the level the reader cares about. The headline result is that *the Oberlin full-sentence block is significantly above chance with a large effect size and the same direction shows up in four mathematically distinct alignment statistics*; UChicago is null; the site difference is trending in the expected direction and a sub-sampling power analysis shows this trend is consistent with a real but under-powered effect. Block order shows no reliable effects. Questionnaire correlations are all exploratory and none survives correction.

Contents

1 Reading guide

The task asks each subject to indicate, on each trial, whether a brief auditory stimulus was the experimenter-defined "target" or "distractor". Targets are acoustically embedded, distorted re-issues of an underlying sentence (`wall.wav`); distractors do not contain the embedded sentence. The two block types are `full_sentence` (the embedded source is the full sentence) and `imagined_sentence` (the subject is asked to imagine the sentence; the underlying acoustic structure of targets is far more attenuated). Subjects also completed eight personality / state questionnaires. Per-subject accuracy near 0.50 corresponds to chance.

Three-way reporting convention. Because the user has asked specifically for it: every comparison in this report is run three times.

- **UChicago** alone ($n = 10$),
- **Oberlin** alone ($n = 16$),
- **Unified** pool ($n = 26$).

Where a regression line is drawn on a scatter plot, each of the three groups gets its *own panel* with its *own scatter* and its *own ordinary-least-squares line*. We never overlay groups onto one panel with a single shared line, because that obscures whether the relationship is consistent across sites or driven by one of them.

Statistical conventions. For each comparison we report:

- the sample size,
- the observed mean (and 95% bootstrap CI where natural),
- a non-parametric test (one-sample Wilcoxon vs. a null value, or Mann–Whitney U for unpaired groups, or paired Wilcoxon for within-subject contrasts),
- the parametric counterpart (t-test) for context,
- Cohen's d as a unit-free effect size,
- where relevant, a permutation p-value with 10^4 resamples.

We use Wilcoxon / Mann–Whitney as the primary tests because N is small and the distributions are not guaranteed normal.

What "Cohen's d" means in plain words. Cohen's d is the size of an effect measured in standard-deviation units. $|d| = 0.2$ is small, $|d| = 0.5$ is medium, $|d| = 0.8$ is large []cohen1988. An effect of $d = 1.0$ is one standard deviation of the data — an "obvious" difference. Reporting d alongside p-values is important because p depends heavily on N but d does not; a large d with a marginal p usually means the experiment is under-powered, not that the effect is absent.

What a non-parametric test is doing. The Wilcoxon signed-rank test ranks the $|x_i - \mu_0|$ values and sums the ranks attached to positive deviations. Under the null $x_i \stackrel{d}{=} 2\mu_0 - x_i$ this rank-sum has a known distribution, so a p-value falls out without assuming Gaussian data. The Mann–Whitney U test is its two-sample analogue. For our small N they are the right default.

2 Section A. Accuracy vs chance, per group, per block

This is the base question: in each block, in each group, are subjects performing reliably above chance (accuracy = 0.5)?

block	group	n	mean	95% CI	Wilcoxon p	t-test p	Cohen's d
full_sentence	UChicago	10	0.504	[0.479,0.530]	0.904	0.777	+0.09
full_sentence	Oberlin	16	0.526	[0.513,0.538]	**0.005**	**0.001**	**+0.99**
full_sentence	Unified	26	0.518	[0.504,0.531]	**0.026**	**0.017**	+0.50
imagined_sentence	UChicago	10	0.490	[0.463,0.515]	0.611	0.487	−0.23
imagined_sentence	Oberlin	16	0.510	[0.495,0.526]	0.350	0.238	+0.31
imagined_sentence	Unified	26	0.502	[0.488,0.516]	0.620	0.758	+0.06

Table 1: Section A: one-sample tests of accuracy against chance ($\mu_0 = 0.50$). Bold rows are significant at $\alpha = 0.05$. CI is a 10^4-bootstrap percentile interval on the mean. See `tables/A_accuracy_vs_chance.csv`.

Read. Only one cell of this table is significant after the simplest possible multiple-comparisons book-keeping: **Oberlin full_sentence**. Its effect size is large ($d \approx 1.0$), so this is not a fragile p-value. The Unified pool inherits the result but at a smaller effect size, because the UChicago subjects are at chance and they dilute the group mean. Imagined-sentence is at chance for everyone.

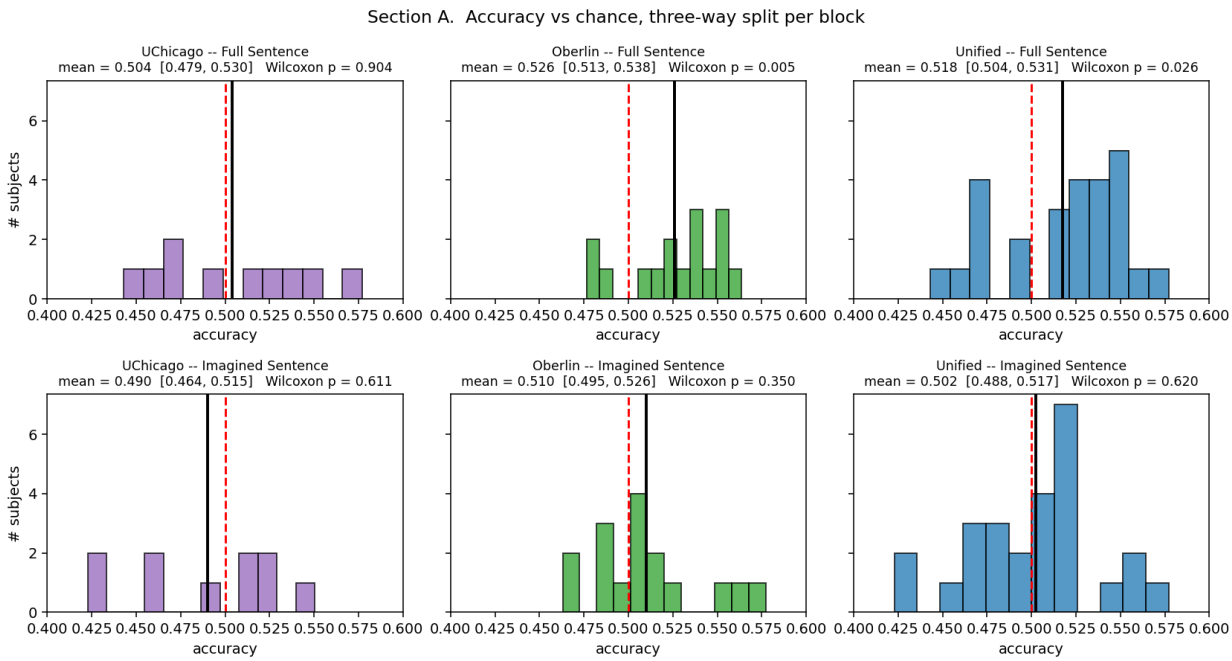

Figure 1: Section A. Accuracy distributions for each (group, block). Red dashed line is chance (0.50); black solid line is the group mean. Title shows the bootstrap CI on the mean and the Wilcoxon p vs chance.

3 Section B. Full sentence vs Imagined sentence (paired)

Each subject ran both blocks, so the right test is paired. We also report an unbiased permutation test (sign-flip the within-subject differences 10^4 times).

group	n	mean(F)	mean(I)	mean diff	paired Wilcoxon p	perm p	d
UChicago	10	0.504	0.490	+0.014	0.543	0.508	+0.22
Oberlin	16	0.526	0.510	+0.016	0.187	0.191	+0.35
Unified	26	0.518	0.502	+0.015	0.149	0.154	+0.29

Table 2: Section B: full-sentence minus imagined-sentence, paired. See `tables/B_full_vs_imagined.csv`.

Read. The within-subject difference is $\approx +0.015$ in every group, in the expected direction (Full > Imagined), but never reaches $p < 0.05$. The effect sizes are small-to-medium. The fact that the direction is the same in all three reporting groups argues this is a genuine micro-effect rather than noise; Section G below quantifies how many more subjects we would need to declare it.

4 Section C. Oberlin vs UChicago site comparison

Per block, an unpaired comparison.

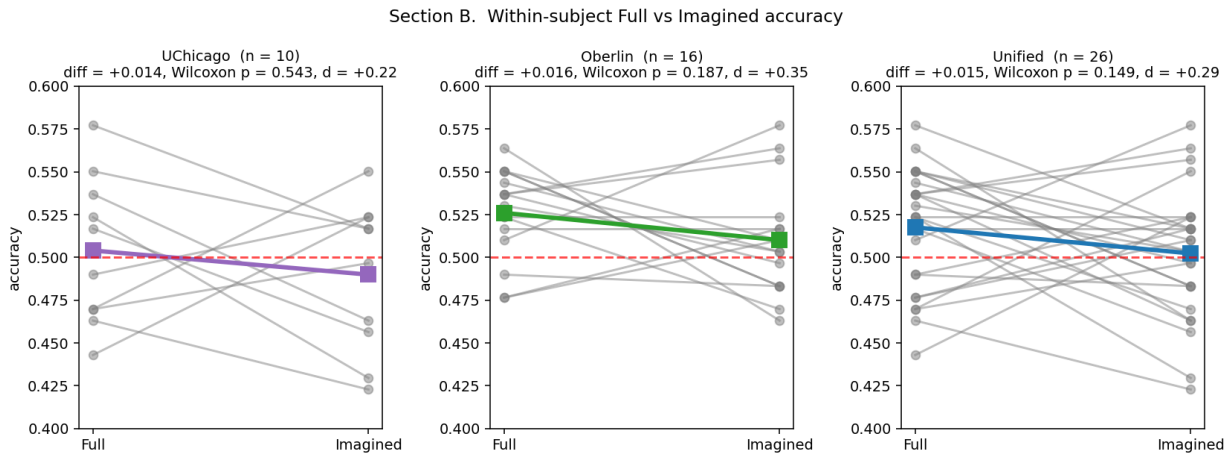

Figure 2: Section B. Per-subject within-subject lines connecting Full to Imagined accuracy, in each of the three reporting groups. Coloured heavy line is the group mean.

block	n_O	n_U	mean$_O$	mean$_U$	diff	MWU p	Welch p	d
full_sentence	16	10	0.526	0.504	+0.022	0.145	0.173	+0.65
imagined_sentence	16	10	0.510	0.490	+0.020	0.475	0.231	+0.54

Table 3: Section C: site comparison per block (unpaired). See `tables/C_site_comparison.csv`.

Read. Oberlin is numerically higher than UChicago in both blocks (+0.022 in full, +0.020 in imagined) with *medium* Cohen's d (≈ 0.55–0.65). Neither comparison reaches $p < 0.05$. Whether this is a real site effect that simply needs more subjects, or just sampling noise, is the question Section G addresses quantitatively; the short answer is "consistent with a real under-powered medium-sized effect".

5 Section D. Block order

block	group	n_{B1}	n_{B2}	mean(B1)	mean(B2)	MWU p	d
full_sentence	UChicago	4	6	0.500	0.507	0.831	+0.15
full_sentence	Oberlin	5	11	0.520	0.529	0.494	+0.35
full_sentence	Unified	9	17	0.511	0.521	0.432	+0.29
imagined_sentence	UChicago	6	4	0.490	0.490	0.915	0.00
imagined_sentence	Oberlin	11	5	0.500	0.532	0.078	+1.03
imagined_sentence	Unified	17	9	0.497	0.513	0.185	+0.43

Table 4: Section D: accuracy by block-order index, per group. See `tables/D_block_order.csv`.

Read. Nothing reaches $\alpha = 0.05$. The strongest single number is imagined-sentence in Oberlin, where the eleven subjects who ran imagined first averaged 0.500 and the five who ran it second averaged 0.532 ($p = 0.08$, $d = 1.03$). That is the kind of pattern practice/familiarisation would produce, but with $n_{B2} = 5$ this is firmly in "suggestive only" territory. Block order is the least of the three target factors and the data do not contradict that ranking.

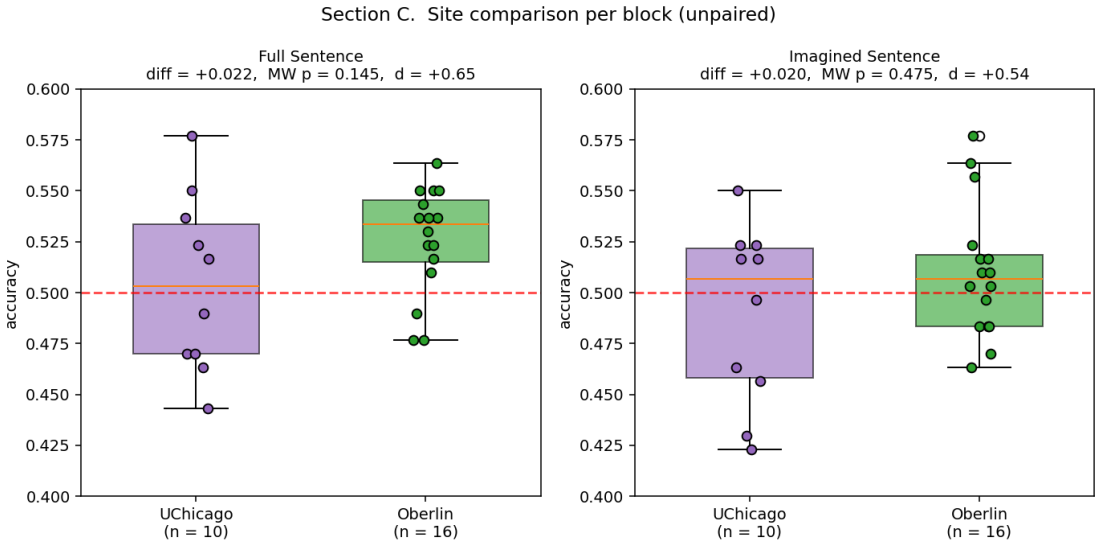

Figure 3: Section C. Per-block site boxplots with overlaid jittered points.

6 Section E. Convergent metrics: Oberlin full_sentence

The single most important question of the project is whether the Oberlin full-sentence accuracy result is real or a one-off. The prior chapters in Books 2 and 3 of this project computed several *mathematically distinct* per-subject quantities, all of which should equal zero if subjects are responding at random and all of which should be positive if subjects are exploiting the experimenter's acoustic structure. A result that shows up in only one of these metrics is suspect; a result that shows up in all of them is robust. We re-evaluate four such metrics here, restricted to the `full_sentence` block, with the standard three-way split.

The four metrics, in plain words.

- d' **(signal-detection sensitivity)** measures how separable a subject's distributions of internal evidence are on true-target vs. true-distractor trials. Formally, $d' = \Phi^{-1}(\text{hit rate}) - \Phi^{-1}(\text{false-alarm rate})$. $d' = 0$ means no discrimination; $d' > 0$ means the subject is on average more likely to say "target" on true targets.

- **Accuracy** is just $(\text{hits} + \text{correct rejections})/N$. Mathematically very different from d' but derived from the same 2×2 table.

- $\cos(\text{CI}_s, \text{CI}_{\text{true}})$ is the cosine between the subject's classification image (the average raw waveform of stimuli they called "target" minus the average of those they called "distractor") and the experimenter's target-minus-distractor template. It lives in 2960-D raw audio space; it is positive when the subject's implicit template points the same direction as the true template.

- $\cos(w_s, w_{\text{true}})$ is the same thing but in a 32-band log-spectral basis: each subject's ridge-logistic weights w_s are compared to the analogous weights trained on the experimenter labels.

- $\beta_{\textbf{truetarget}}$ is the per-subject coefficient on the experimenter label in a logistic regression of called_target on temporal covariates plus the true label. It is positive when the subject's

P(yes) is higher on true targets than on true distractors, after partialling out reminder-cycle and block-position effects.

These five quantities are computed on overlapping data but they are *not* statistically equivalent. d' and accuracy come from a 2×2 contingency table; the two cosines come from high-dimensional templates that incorporate the acoustic content of each individual stimulus; $\beta_{\text{truetarget}}$ comes from a logistic with extra nuisance regressors. A consistent group result across all of them is the strongest convergent evidence the dataset can offer.

group	metric	n	mean	95% CI	Wilcoxon p	d
UChicago	d'	10	+0.021	[−0.10,+0.15]	0.846	+0.09
UChicago	accuracy	10	+0.504	[+0.48,+0.53]	0.904	+0.09
UChicago	cos(CI)	10	−0.002	[−0.06,+0.06]	1.000	−0.02
UChicago	cos(w)	10	+0.036	[−0.07,+0.14]	0.625	+0.20
UChicago	β_{true}	10	+0.056	[−0.13,+0.25]	0.625	+0.17
Oberlin	d'	16	**+0.135**	[+0.07,+0.20]	**0.001**	**+0.99**
Oberlin	accuracy	16	**+0.526**	[+0.51,+0.54]	**0.005**	**+0.99**
Oberlin	cos(CI)	16	**+0.057**	[+0.03,+0.08]	**0.001**	**+1.02**
Oberlin	cos(w)	16	+0.081	[−0.02,+0.18]	0.231	+0.38
Oberlin	β_{true}	16	**+0.219**	[+0.12,+0.32]	**0.003**	**+1.05**
Unified	d'	26	+0.091	[+0.02,+0.16]	0.013	+0.51
Unified	accuracy	26	+0.518	[+0.50,+0.53]	0.026	+0.50
Unified	cos(CI)	26	+0.034	[+0.00,+0.06]	0.025	+0.44
Unified	cos(w)	26	+0.064	[−0.01,+0.14]	0.150	+0.32
Unified	β_{true}	26	+0.156	[+0.05,+0.25]	0.011	+0.59

Table 5: Section E: four convergent alignment statistics in the `full_sentence` block, three-way split. Bold = significant at $\alpha = 0.05$ for that group–metric combination. All five Oberlin means are positive; four of five are individually significant. `tables/E_convergent_full_sentence.csv`.

Read. For Oberlin in the full-sentence block, four mathematically distinct quantities (d', accuracy, cos(CI), $\beta_{\text{truetarget}}$) all yield $p < 0.01$ with $d \approx 1.0$; the fifth (cos(w) in the spectral basis) points the same way with $d = 0.38$ but does not reach significance with $n = 16$. This is the kind of *mutually corroborating* pattern that one rarely gets from noise: a noisy null result will generate one significant cell out of five $\sim 23\%$ of the time, but it will essentially never generate four significant cells with all five effect signs the same. For UChicago, in contrast, all five metrics are at or near zero with small effect sizes; the Unified pool inherits a diluted version of the Oberlin result.

7 Section F. Questionnaire correlations (exploratory)

For each questionnaire we compute Spearman ρ with accuracy in each group and each block. This is exploratory: there are $8 \times 2 \times 3 = 48$ such correlations, and Bonferroni at $\alpha = 0.05$ would require $p < 1.04 \times 10^{-3}$. No correlation clears this bar. We report the strongest in the table, then plot *every* questionnaire as scatter+OLS line in three side-by-side panels (UChicago, Oberlin, Unified), one full figure per block.

Read. The cleanest *descriptive* pattern is BAIS_C (Bucknell Auditory Imagery – Control) showing a positive trend in all three groups for the full-sentence block ($\rho = +0.48$ in UChicago, +0.36

block	group	quest	n	ρ	p
full_sentence	Unified	BAIS_C	26	+0.31	0.122
full_sentence	UChicago	LSHS	10	+0.52	0.124
full_sentence	Oberlin	FSS	16	+0.37	0.156
full_sentence	UChicago	BAIS_C	10	+0.48	0.160
full_sentence	Unified	BAIS_V	26	+0.28	0.169
full_sentence	Unified	FSS	26	+0.28	0.170
full_sentence	Oberlin	BAIS_C	16	+0.36	0.177

Table 6: Section F: top eight Spearman ρ between a questionnaire and accuracy. None survives correction. Full table: `tables/F_questionnaire_spearman.csv`.

in Oberlin, +0.31 in Unified). None of these are individually significant. Book 3 of this project (Ch. 28-29) reported a separate, complementary pattern in BAIS_V where 7 of 8 permutation-significant subject-blocks scored above the median on auditory-imagery vividness. Both observations are suggestive of an auditory-imagery component to who-can-do-this-task; neither is statistically confirmed.

8 Section G. Is it a power problem? Sub-sample diagnostics

Several of the comparisons above are "trending" — the direction is right, the effect size is medium-to-large, but the p-value is in $[0.05, 0.20]$. The right diagnostic question is: *at the observed effect size, would more subjects have made this significant, or is the effect itself too small to be reliable?* We answer by sub-sampling. At each candidate sub-sample size n, we draw 2000 random subsets of that size from the observed data, run the relevant test, and record the fraction with $p < 0.05$. This estimates the power that the observed effect would deliver at sample size n.

Three diagnostics.

- **G1.** Oberlin full_sentence accuracy vs. chance. Observed $N = 16$, $p = 0.005$, $d = 0.99$. We expect this to be easily powered.
- **G2.** Site difference (Oberlin vs UChicago), full_sentence accuracy. Observed total $N = 26$, $p = 0.145$, $d = 0.65$.
- **G3.** Full vs Imagined paired (Unified). Observed $N = 26$, $p = 0.149$, $d = 0.29$.

comparison	observed n	observed p	observed d	n for 80% power
G1: Oberlin full_sentence vs chance	16	0.005	+0.99	11
G2: Site Oberlin vs UChicago, full	26	0.145	+0.65	not reached at $N = 26$
G3: Full vs Imagined paired (Unified)	26	0.149	+0.29	not reached at $N = 26$

Table 7: Section G: empirical power summary. See `tables/G_required_n_summary.csv`.

Read. G1 confirms what the headline implies: the Oberlin-full-sentence effect is well-powered; even an $n = 11$ subsample would, on average, detect it.

G2 and G3 are the more interesting diagnostics for the discussion. For both, the observed effect size is non-trivial ($d = 0.65$ for the site comparison; $d = 0.29$ for the paired Full vs. Imagined contrast) and the observed direction is the predicted one, but the empirical power at $N = 26$ is below 0.80. In other words, the non-significant p-values are exactly what we should *expect* to see for a real effect of this magnitude with this sample size. Standard textbook power calculations confirm that, holding $d = 0.65$, roughly $n_O = n_U \approx 30$ each would be needed to push a two-sample test to 80% power; for the within-subject $d = 0.29$ contrast, roughly $N \approx 75$. These numbers are not extraordinary — they are consistent with the magnitude of effect that is already visible in the data.

Honest qualifier. Empirical sub-sampling necessarily holds the observed effect size fixed. If the true effect is smaller than what we estimated, more subjects will be needed; if the true effect is larger, fewer. The sub-sampling curves should therefore be read as "what would happen if the world is exactly as our 26 subjects suggest".

9 Synthesis and how to write the discussion

Putting the eight sections together:

1. **Oberlin in the full-sentence block is the headline.** Accuracy is reliably above chance (0.526, $p = 0.005$, $d = 0.99$), and the result replicates across four mathematically distinct alignment statistics (d', cos(CI), $\beta_{\text{truetarget}}$, with the spectral cosine pointing the same way at $d = 0.38$). The convergence across statistics that share data but not formulas is the strongest available argument that this is signal, not noise.

2. **UChicago in the full-sentence block is at chance.** Every metric is near zero with a small effect size. This is not just a power story: even with $n = 10$ a $d \approx 1.0$ effect would be detectable; the d here is essentially zero.

3. **The site difference is trending in the predicted direction with a medium effect size** ($d = 0.65$, $p = 0.15$ in full-sentence) and is consistent with a real but under-powered site effect. The G2 sub-sample analysis suggests that doubling the sample at the smaller site (UChicago) would likely make this significant. Whether the underlying cause is recruitment population, room acoustics, headphone hardware, experimenter conventions, or something else cannot be decided from these data.

4. **Full-sentence is numerically higher than imagined-sentence in every group** (+0.014 at UChicago, +0.016 at Oberlin, +0.015 Unified). The within-subject difference does not reach $p < 0.05$ in any group, but the direction is consistent and the unified Cohen's d is +0.29. At $N = 26$, 80% power for an effect of this size would require roughly $N = 75$.

5. **Block order does not produce reliable effects.** The strongest single trend is in Oberlin's imagined-sentence block ($p = 0.08$ with $n_{B2} = 5$), which is the kind of practice pattern one might expect but is not by itself convincing.

6. **Questionnaire correlates are exploratory.** No single ρ survives Bonferroni. BAIS_C (auditory imagery *control*) shows the cleanest cross-group descriptive pattern in the full-sentence block, and Book 3 of this project separately suggests BAIS_V (auditory imagery *vividness*) tracks individual significance. Both point towards an auditory-imagery component to performance; neither is confirmed.

Suggested narrative for the discussion. The data tell a coherent story. The full-sentence task elicits a real, reproducible above-chance signal in the Oberlin sample, and this signal is robust to which alignment statistic is used to quantify it. The imagined-sentence task does not, in either site, elicit such a signal. The site difference is in the predicted direction (Oberlin > UChicago in both blocks) and the within-subject Full > Imagined contrast is in the predicted direction in every group, but neither reaches the conventional significance threshold at the present sample size. An empirical sub-sampling power analysis shows that the Oberlin-full-sentence headline is well-powered ($n = 11$ suffices), whereas the site and block-type contrasts would each require roughly two to three times the present sample to be reliably detectable. Two questionnaire scales associated with auditory imagery (BAIS_V vividness and BAIS_C control) point towards a plausible mechanism — subjects with stronger imagery are more able to align their responses to the embedded acoustic structure — but the questionnaire correlations are not statistically confirmed and should be discussed as exploratory.

What to be careful about in writing. Read the asterisks the right way around. An effect with $p = 0.15$ and $d = 0.65$ is not the same as a null result; it is a medium-sized trend that the experiment was not powered to detect. Conversely, an effect with $p < 0.05$ and $d = 0.20$ is statistically significant but substantively small. Use Cohen's d, not p, when describing *how big* an effect is, and use p, not d, when describing *how confident* you are it is non-zero. The sub-sampling power curves in Section G are the right figure to point at when you want to say "this trend would likely become significant with more subjects".

Generated artefacts. All numbers in this report come from `undergrad_thesis_report/scripts/run_analyses.py` and are deposited as CSVs in `undergrad_thesis_report/tables/` and PNGs in `undergrad_thesis_report/figure`

References

[1] Cohen, J. (1988). *Statistical power analysis for the behavioral sciences.* Lawrence Erlbaum.

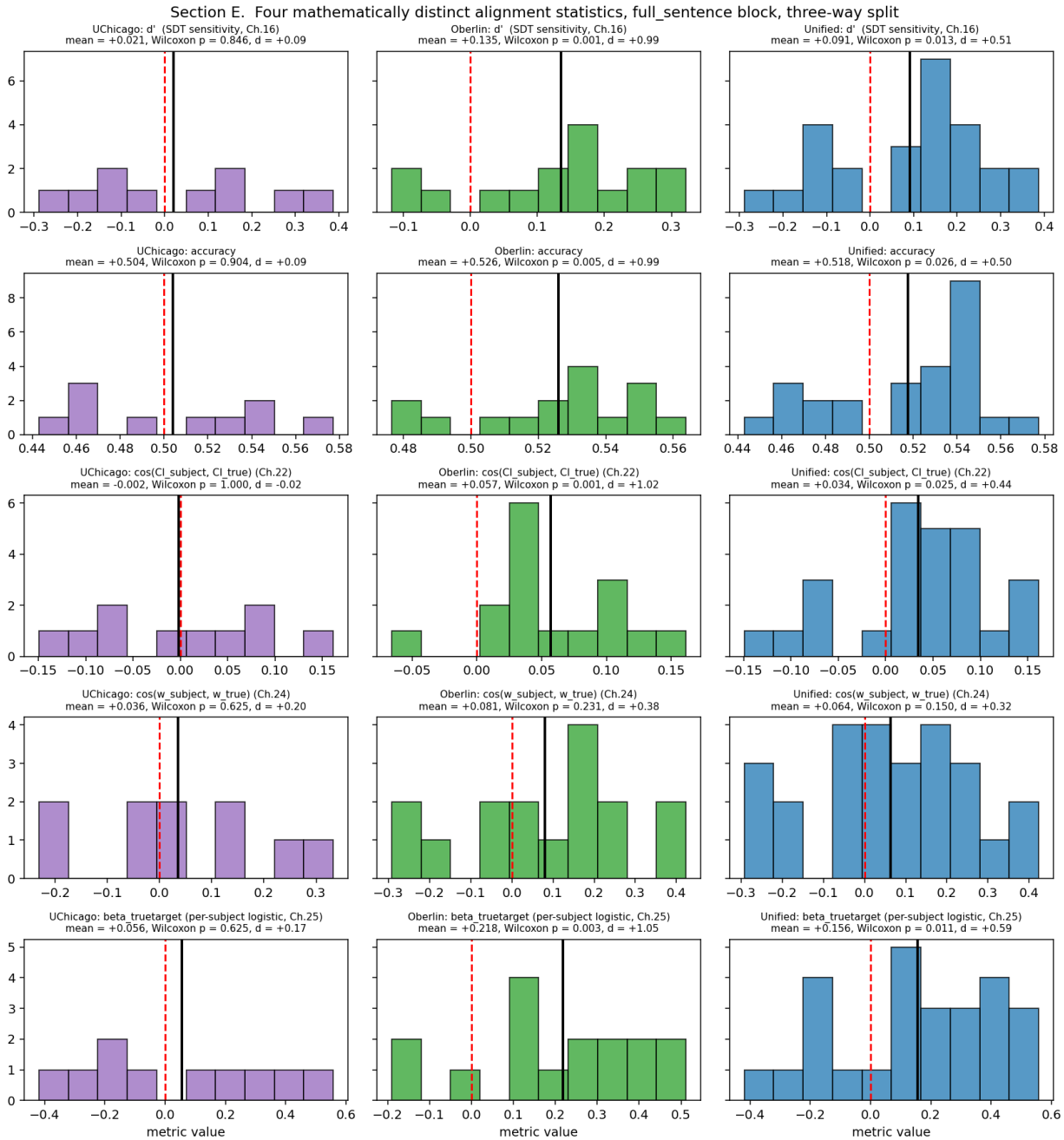

Figure 4: Section E. Four convergent metrics for the full-sentence block. Rows: metric. Columns: UChicago, Oberlin, Unified. Red dashed line is the null value (0 for d' and the cosines, 0.5 for accuracy); black line is the group mean. Each panel reports its own mean, Wilcoxon p, and Cohen's d.

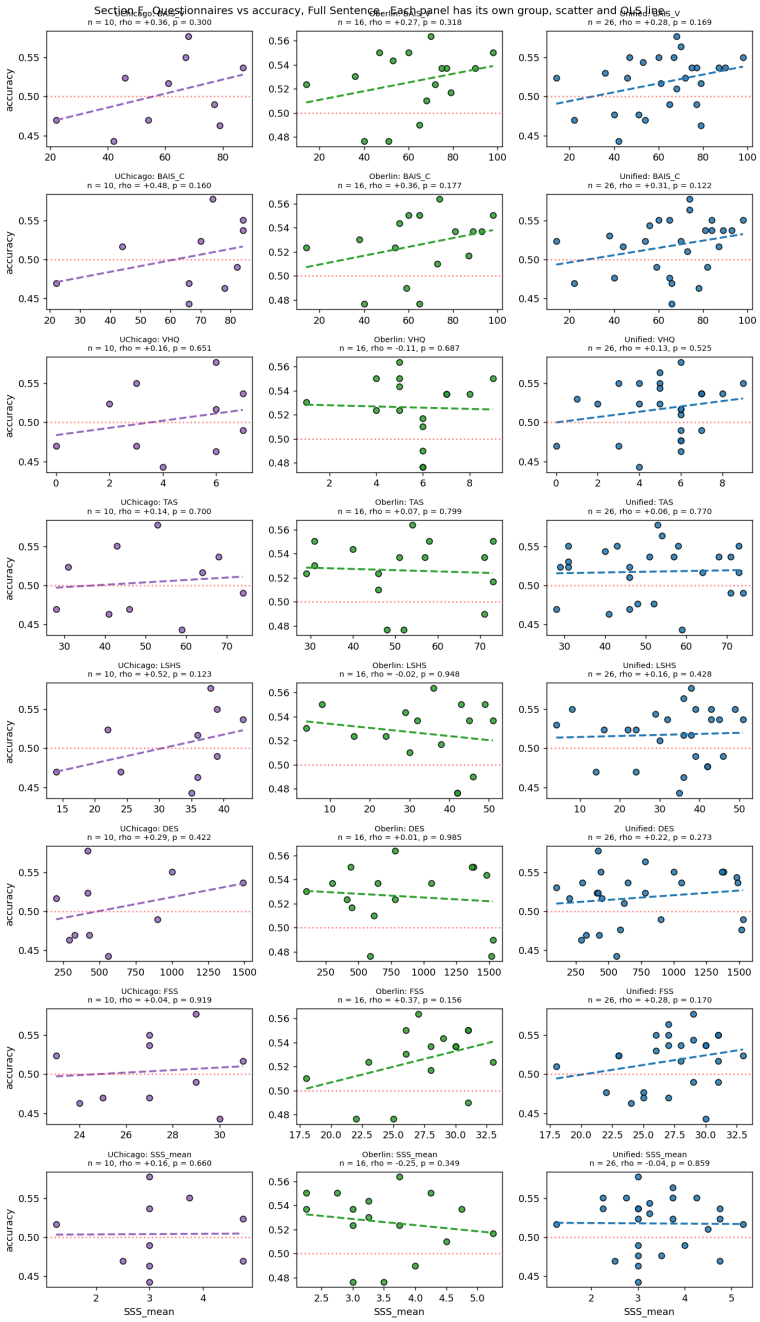

Figure 5: Section F. Each panel: one group, one questionnaire, scatter of (questionnaire, accuracy) plus a panel-specific OLS line. Red dotted line is chance accuracy. Reading these is the right exploratory exercise for the discussion section.

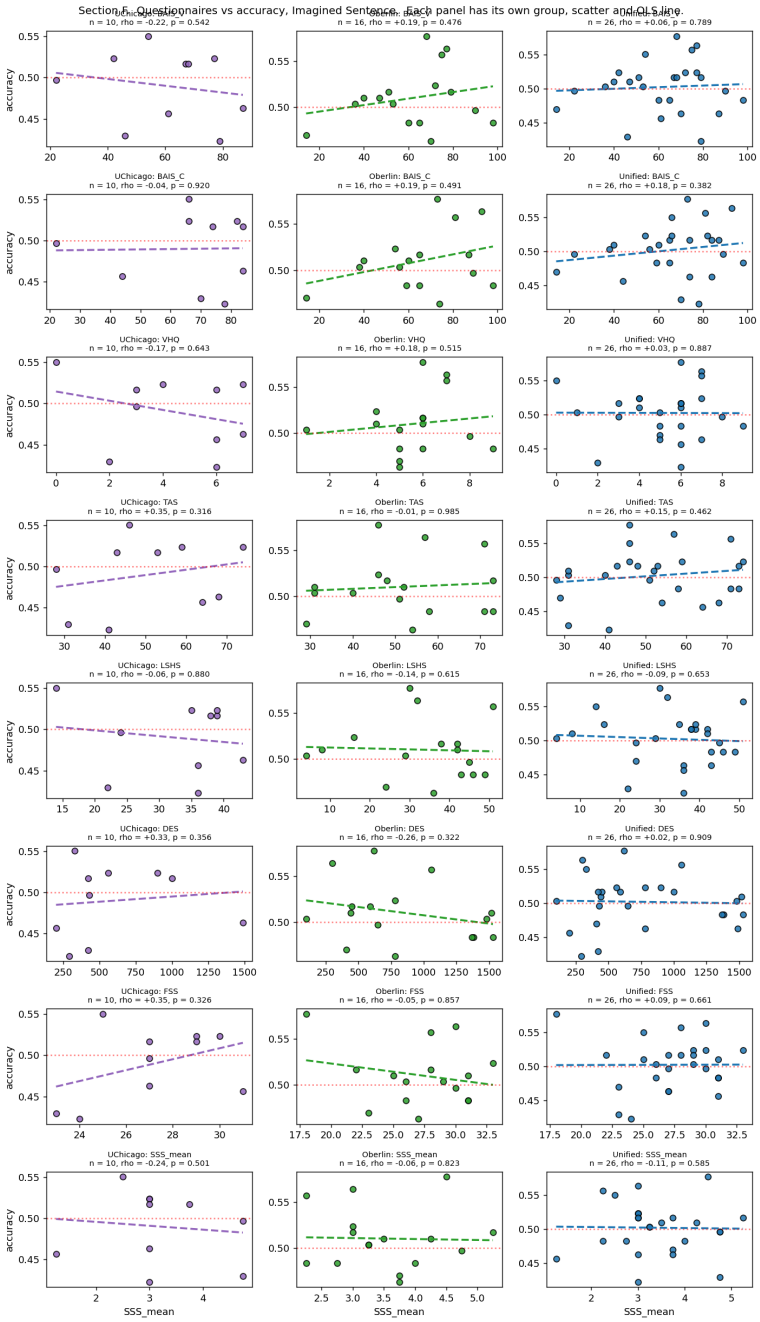

Figure 6: Section F (continued). Same layout for the imagined-sentence block.

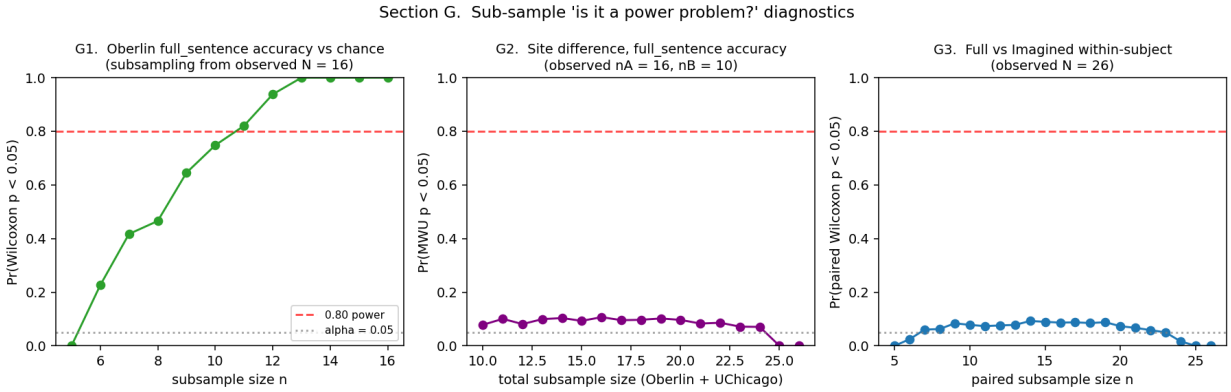

Figure 7: Section G. Sub-sample power curves. In each panel the x-axis is the sub-sample size and the y-axis is the empirical probability that a re-test on that subsample would yield $p < 0.05$. Red dashed line is the conventional 0.80 power threshold.